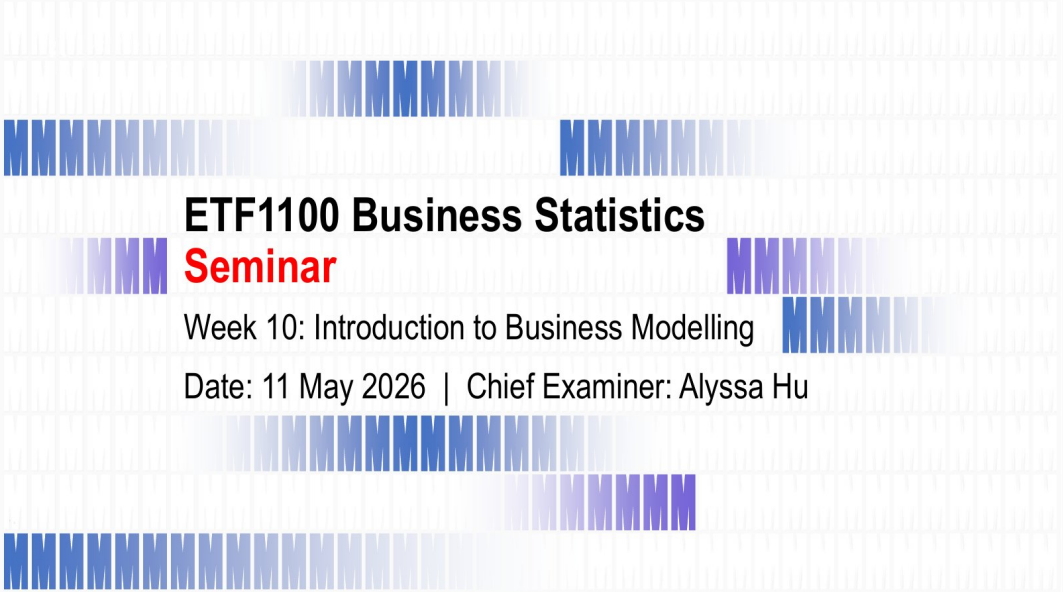


Figure 1: ETF1100 Week 10 Seminar source PDF, page 1

Reminders/FAQs

Please make sure you are reading unit announcements!

Weekly recordings and assignments: Moodle > Learning tab > Week 10

How can I ask questions outside of class time?

- All Day TA chatbot
- Weekly Q&A Discussion Forum

Moodle > Learning > Week 10 > Overview

- Join student consultation hours

Moodle > Unit Information

- ETF1100.Caulfield-x@monash.edu for personal/sensitive communication

Figure 2: ETF1100 Week 10 Seminar source PDF, page 2

Case Study Statistics

- Some students requested summary statistics for the case study:
  - Mean: 88.34%, with standard deviation 13.23%
  - Median: 92.50%
  - Keep up the great work!

Figure 3: ETF1100 Week 10 Seminar source PDF, page 3

Overview

- Last week, we continued learning about the multiple linear regression model (MLR). This included testing for significant coefficients and overall model fit, evaluating models, and handling categorical variables.
- This week, we will focus on spreadsheet-based business modelling, which can help us solve optimisation problems.
- We will use the Week 10 Seminar Data.

Figure 4: ETF1100 Week 10 Seminar source PDF, page 4

Learning Outcomes

- Developing skills in spreadsheet modeling of business problems
- Solving optimisation problems in a business setting using Excel
- Exploring methods for sensitivity analysis
- Developing an understanding for how simulation can be used to address business problems

Figure 5: ETF1100 Week 10 Seminar source PDF, page 5

Motivation

Spreadsheet-based business modelling techniques have many applications in business and economics:

- Determining optimal price of product
- Balancing marketing mix
- Investing decisions
- Identifying production level at different manufacturing plants
- Simplifying logistics with location and number of warehouses

Figure 6: ETF1100 Week 10 Seminar source PDF, page 6

What is optimisation?

Process of finding the best decisions for a particular measure of performance

In defining “best decision” and “performance”, we need to have:

- Known input parameters
- A set of decisions – things that the business can control
- Some constraints on the decision
- An objective or criteria to reach

These can be formulated to a model.

Figure 7: ETF1100 Week 10 Seminar source PDF, page 7

Example:

A business firm wants to determine a selling price for Product A. The unit cost of Product A is known to be $40 and the demand curve is known to be $y = 800 - 5x$.

- x: selling price per unit of Product A
- y: quarterly demand for Product A

Thus,

- Known input parameters: 40 (unit cost), -5 (marginal effect of price on demand), 800 (maximum demand)
- A set of decisions – things that the business can control: x
- Some constraints on the decision: $x \geq 40$, so that profit $z \geq 0$
- An objective or criteria: maximize quarterly profit $z = (x - 40)y$

Figure 8: ETF1100 Week 10 Seminar source PDF, page 8

How can we use Excel to model this?

1. **Excel's Data Table Function**
   It explores how the objective changes if decisions or input parameters change. Returns table of calculations/results.
2. **Excel's Solver**
   It runs constrained optimisation that solves a maximization or minimization problem subject to constraints. Returns best feasible solution.

Figure 9: ETF1100 Week 10 Seminar source PDF, page 9

How can we use Excel to model this?

Setting up the mathematical formula in Excel:

- Inputs
- Decision
- Necessary Calculations
- Objective

	A	B	C	D	E	F	G
1	Price, Demand, and Profit						
2							
3	Inputs						
4		Max Demand	800				
5		Slope	-5				
6		Unit Cost	40				
7							
8	Decision						
9		Unit Price	80	<-- Try x = 80 to start; we can modify/control this			
10							
11	Calculation						
12		Demand	400	<-- Set to = C4+C5*C9			
13							
14	Objective						
15		Profit	16000	<-- Set to = (C9-C6)*C12			

See demo and recording.

Figure 10: ETF1100 Week 10 Seminar source PDF, page 10

Data Table

Next, let's try different prices using Data Table:

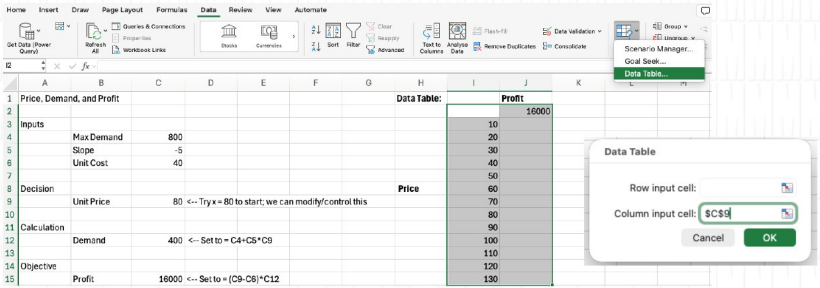
	A	B	C	D	E	F	G	H	I	J
1	Price, Demand, and Profit							Data Table:		Profit
2										=C15
3	Inputs								10	
4		Max Demand	800						20	
5		Slope	-5						30	
6		Unit Cost	40						40	
7									50	
8	Decision							Price	60	
9		Unit Price	80	<-- Try x = 80 to start; we can modify/control this					70	
10									80	
11	Calculation								90	
12		Demand	400	<-- Set to = C4+C5*C9					100	
13									110	
14	Objective								120	
15		Profit	16000	<-- Set to = (C9-C6)*C12					130	

See demo and recording.

Figure 11: ETF1100 Week 10 Seminar source PDF, page 11

Data Table

Highlight your data table (without the heading labels), then go to Data > What-if Analysis > Data Table



See demo and recording.

Figure 12: ETF1100 Week 10 Seminar source PDF, page 12

Data Table

Output:



Data Table:		Profit
		16000
	10	-22500
	20	-14000
	30	-6500
	40	0
	50	5500
Price	60	10000
	70	13500
	80	16000
	90	17500
	100	18000
	110	17500
	120	16000
	130	13500

- Pricing below $40 generate loss
- Profit is non-linearly related to price
- Over this crude grid, the maximum profit of $18,000 is achieved at price $100 per unit

See demo and recording.

Figure 13: ETF1100 Week 10 Seminar source PDF, page 13

Data Table

Next, let's try different costs as well:

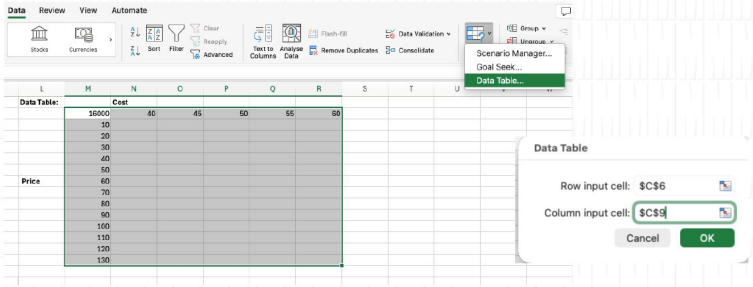
L	M	N	O	P	Q	R
Data Table:		**Cost**				
	= C15	40	45	50	55	60
	10					
	20					
	30					
	40					
	50					
Price	60					
	70					
	80					
	90					
	100					
	110					
	120					
	130					

See demo and recording.

Figure 14: ETF1100 Week 10 Seminar source PDF, page 14

Data Table

Highlight your data table (without the heading labels), then go to Data > What-if Analysis > Data Table



See demo and recording.

Figure 15: ETF1100 Week 10 Seminar source PDF, page 15

Data Table

Output:

Data Table:	Cost					
	16000	40	45	50	55	60
	10	-22500	-26250	-30000	-33750	-37500
	20	-14000	-17500	-21000	-24500	-28000
	30	-6500	-9750	-13000	-16250	-19500
	40	0	-3000	-6000	-9000	-12000
	50	5500	2750	0	-2750	-5500
Price	60	10000	7500	5000	2500	0
	70	13500	11250	9000	6750	4500
	80	16000	14000	12000	10000	8000
	90	17500	15750	14000	12250	10500
	100	18000	16500	15000	13500	12000
	110	17500	16250	15000	13750	12500
	120	16000	15000	14000	13000	12000
	130	13500	12750	12000	11250	10500

- Optimal price of $100/unit for cost of $40, $45, and $50
- Optimal price of $110/unit for cost of $55 and $60
- Note that as cost goes up, the maximum profit at the optimal price decreases.

See demo and recording.

Figure 16: ETF1100 Week 10 Seminar source PDF, page 16

Solver

We note that if we keep increasing the possible prices and costs, the table can become unwieldy. Instead of having to write out all possible values, we can call on Solver to get an optimal solution under constraint.

Data > Analysis Tools > Check “Solver Add-in”

See demo and recording.

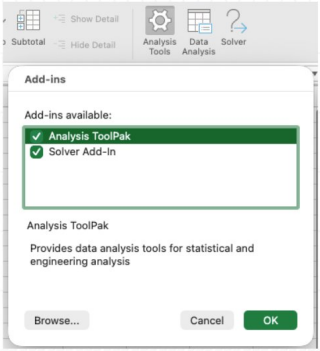
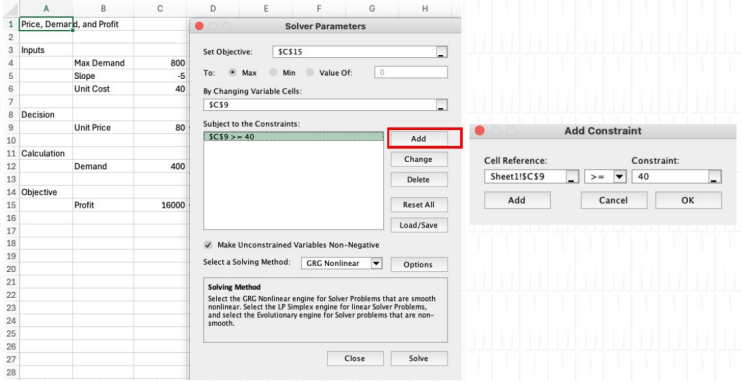


Figure 17: ETF1100 Week 10 Seminar source PDF, page 17

Solver

Using Solver:

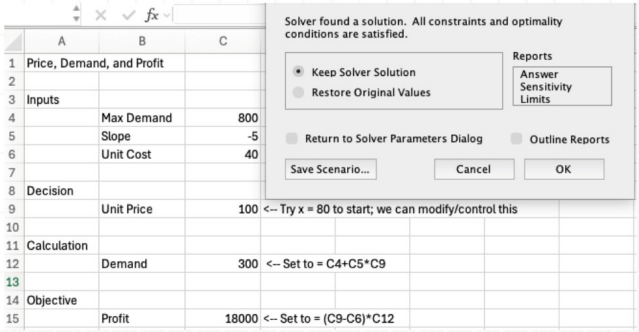


See demo and recording.

Figure 18: ETF1100 Week 10 Seminar source PDF, page 18

Solver

Using Solver



	A	B	C	
1	Price, Demand, and Profit			
2				
3	Inputs			
4		Max Demand	800	
5		Slope	-5	
6		Unit Cost	40	
7				
8	Decision			
9		Unit Price	100	<-- Try x = 80 to start; we can modify/control this
10				
11	Calculation			
12		Demand	300	<-- Set to = C4+C5*C9
13				
14	Objective			
15		Profit	18000	<-- Set to = (C9-C6)*C12

See demo and recording.

Figure 19: ETF1100 Week 10 Seminar source PDF, page 19

Example: Marketing Mix

Suppose the owner of a pizza company wants to figure out how much to spend on advertising through pamphlets.



We are given the following information:

- Profit = (Price x Quantity) - Total Cost
- Total Cost = Variable Cost + Fixed Cost
- Fixed Cost = 700 + Pamphlet Price x Number of Pamphlets
- Variable Cost = (Wage/5 + Cost of Ingredients) x Pizza Quantity
- Pizza Quantity = 120 + (0.2 x Number of Pamphlets) – (0.0003 x Number of Pamphlets^2)
- Price = 25, Wage = 20, Pamphlet Price = 2, Cost of Ingredients = 3

Figure 20: ETF1100 Week 10 Seminar source PDF, page 20

Example: Marketing Mix

Let's get this into our spreadsheet. Formulas included for reference.

	A	B	C	D	E
1	Pizza				
2					
3	Inputs				
4		Pizza Price	25		
5		Wage	20		
6		Pamphlet Price	2		
7		Ingredient Cost	3		
8					
9	Decision				
10		Pamphlet Number	100	<- try 100 to start	
11					
12	Calculation				
13		Pizza Quantity	137		
14		Fixed Cost	900		
15		Variable Cost	959		
16		Total Cost	1859		
17					
18	Objective				
19		Profit	1566		

= 120 + 0.2 * C10 - 0.0003 *C10^2
= 700 + C6*C10
= (C5/5 + C8)*C13
= C14 + C15
= (C4*C13) - C16

See demo and recording.

Figure 21: ETF1100 Week 10 Seminar source PDF, page 21

Example: Marketing Mix – Data Table

Suppose we vary both pamphlet number and price. Using the cells in this screenshot, which cell is the row input cell? Column input cell?

G2 =C19

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Pizza						Data Table:			Pamphlet Price			
2							1566	0.5	1	1.5	2	2.5	3
3	Inputs						50						
4		Pizza Price	25				100						
5		Wage	20			# of Pamphlet	150						
6		Pamphlet Price	2				200						
7		Ingredient Cost	3				250						
8							300						
9	Decision												
10		Pamphlet Number	100	<- try 100 to start									
11													
12	Calculation												
13		Pizza Quantity	137										
14		Fixed Cost	900										
15		Variable Cost	959										
16		Total Cost	1859										
17													
18	Objective												
19		Profit	1566										

See demo and recording.

PollEv.com/alyssahu

Figure 22: ETF1100 Week 10 Seminar source PDF, page 22

Example: Marketing Mix – Data Table

This is our final data table:

	Data Table:			Pamphlet Price			
	1566	0.5	1	1.5	2	2.5	3
	50	$ 1,601.50	$ 1,576.50	$ 1,551.50	$ 1,526.50	$ 1,501.50	$ 1,476.50
	100	$ 1,716.00	$ 1,666.00	$ 1,616.00	$ 1,566.00	$ 1,516.00	$ 1,466.00
# of Pamphlet	150	$ 1,803.50	$ 1,728.50	$ 1,653.50	$ 1,578.50	$ 1,503.50	$ 1,428.50
	200	$ 1,864.00	$ 1,764.00	$ 1,664.00	$ 1,564.00	$ 1,464.00	$ 1,364.00
	250	$ 1,897.50	$ 1,772.50	$ 1,647.50	$ 1,522.50	$ 1,397.50	$ 1,272.50
	300	$ 1,904.00	$ 1,754.00	$ 1,604.00	$ 1,454.00	$ 1,304.00	$ 1,154.00

- Lower pamphlet price → lower fixed cost → optimal profit is higher.
- At $2 per pamphlet, the profit maximising number of pamphlets is 150. At $0.5 per pamphlet, the optimal number of pamphlets is 300.

See demo and recording.

Figure 23: ETF1100 Week 10 Seminar source PDF, page 23

Example: Marketing Mix

Suppose the pizza company can advertise on Facebook as well as via pamphlets. Facebook ads are more expensive than pamphlets (see fixed cost) but tends to sell more pizza (see pizza quantity).

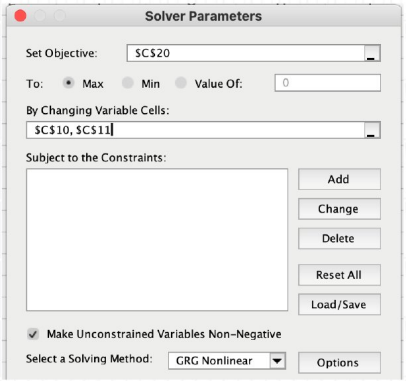
We are given the following updated information:

- Fixed Cost = 700 + (Pamphlet Price x Number of Pamphlets) + (Facebook Price x Number of Facebook Ads)
- Pizza Quantity = 120 + (0.2 x Number of Pamphlets) – (0.0003 x Number of Pamphlets^2) + (0.3 x Number of Facebook Ads) – (0.0003 x Number of Facebook Ads^2)
- Facebook Ad Price = 5

What if we use both Facebook ads and pamphlets?

Figure 24: ETF1100 Week 10 Seminar source PDF, page 24

Example: Marketing Mix - Solver



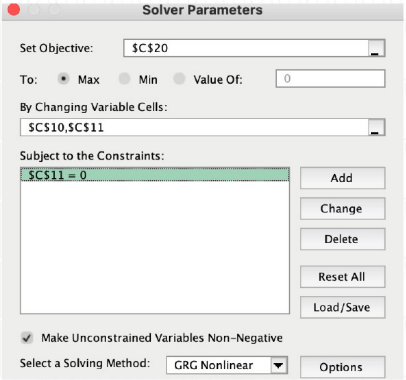
See demo and recording.

	A	B	C	D	E
1	Pizza				
2					
3	Inputs				
4		Pizza Price	25		
5		Wage	20		
6		Pamphlet Price	2		
7		Ingredient Cost	3		
8		Facebook Ad Price	5		
9					
10	Decision	Pamphlet Number	148.1480375	<- try 100 to start	
11		Facebook Ad Number	37.0369524	<- try 100 to start	
12					
13	Calculation				
14		Pizza Quantity	153.7448202		
15		Fixed Cost	1181.480837		
16		Variable Cost	1076.213741		
17		Total Cost	2257.694578		
18					
19	Objective				
20		Profit	1585.925926		

Figure 25: ETF1100 Week 10 Seminar source PDF, page 25

Example: Marketing Mix - Solver

What happens if we get rid of the Facebook ads? Only lose about $7!



See demo and recording.

	A	B	C	D	E
1	Pizza				
2					
3	Inputs				
4		Pizza Price	25		
5		Wage	20		
6		Pamphlet Price	2		
7		Ingredient Cost	3		
8		Facebook Ad Price	5		
9					
10	Decision	Pamphlet Number	148.1481464	<- try 100 to start	
11		Facebook Ad Number	0	<- try 100 to start	
12					
13	Calculation				
14		Pizza Quantity	143.0452673		
15		Fixed Cost	996.2962928		
16		Variable Cost	1001.316871		
17		Total Cost	1997.613164		
18					
19	Objective				
20		Profit	1578.518519		

Figure 26: ETF1100 Week 10 Seminar source PDF, page 26

Example: Investment Decisions

An investor considers purchasing a café. He presents you with the following timeline and the net cashflow (expected profit) per year:

- Year 0: Purchase café for $650000.
- Year 1: Cashflow around $40000.
- Cashflow will increase by 4% per year after that.
- Year 10: Sell café for $800000.

See demo and recording.

	A	B
1	**Year**	**Cashflow**
2	0	-650000
3	1	40000
4	2	=B3*1.04
5	3	=B4*1.04
6	4	=B5*1.04
7	5	=B6*1.04
8	6	=B7*1.04
9	7	=B8*1.04
10	8	=B9*1.04
11	9	=B10*1.04
12	10	800000

Figure 27: ETF1100 Week 10 Seminar source PDF, page 27

Example: Investment Decisions

Is this a good investment or a "lemon"? Let's look at net present value (NPV) and internal rate of return (IRR) to compare investments.

NPV accounts for "time value" of money (i.e. inflation, lost opportunity).

$$NPV = \sum_{t=0}^{10} \frac{C_t}{(1+r)^t}$$

C_t: cashflow in period t

r: discount rate (return that could be earned on alternate investment)

Example of Excel commands:

14	r	0.09
15	NPV	=NPV(B14,B2:B12)
16	IRR	=IRR(B2:B12)

IRR is the discount rate that makes NPV equal to 0 (that is, the projected discounted cashflows equal to the initial investment amount).

Figure 28: ETF1100 Week 10 Seminar source PDF, page 28

Example: Investment Decisions – GoalSeek

- The investor is currently considering another investment opportunity which has an IRR of 9%.
- If the purchase price of $650,000 is negotiable, what price would yield an IRR of at least 9%? (That is, the price such that r = 9% and NPV = 0.)
- We will use our final Excel function 'Goal Seek' which finds the input value needed to hit a target output value.



Figure 29: ETF1100 Week 10 Seminar source PDF, page 29

Example: Investment Decisions – GoalSeek

Data > What-if Analysis > Goal Seek

'Set cell': NPV cell

'To value': 0

'By changing cell': Year 0 cashflow

Goal Seek returns a year 0 cashflow of $613,663. So, we should try to negotiate the price down to $613,663 to make this investment worthwhile.

See demo and recording.

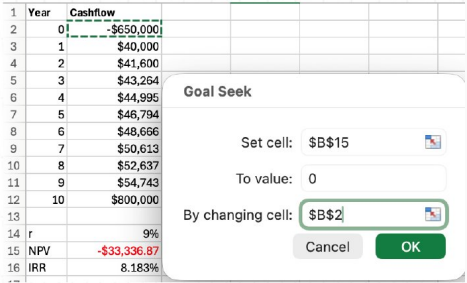


Figure 30: ETF1100 Week 10 Seminar source PDF, page 30

Summarising Week 10 Seminar

- Spreadsheet modelling
  - Ideas to solve business problems through simulation, sensitivity analysis and optimization using Excel
- Excel tools for simulation and optimization
  - Data Table
  - Solver
  - Goal Seek

Figure 31: ETF1100 Week 10 Seminar source PDF, page 31

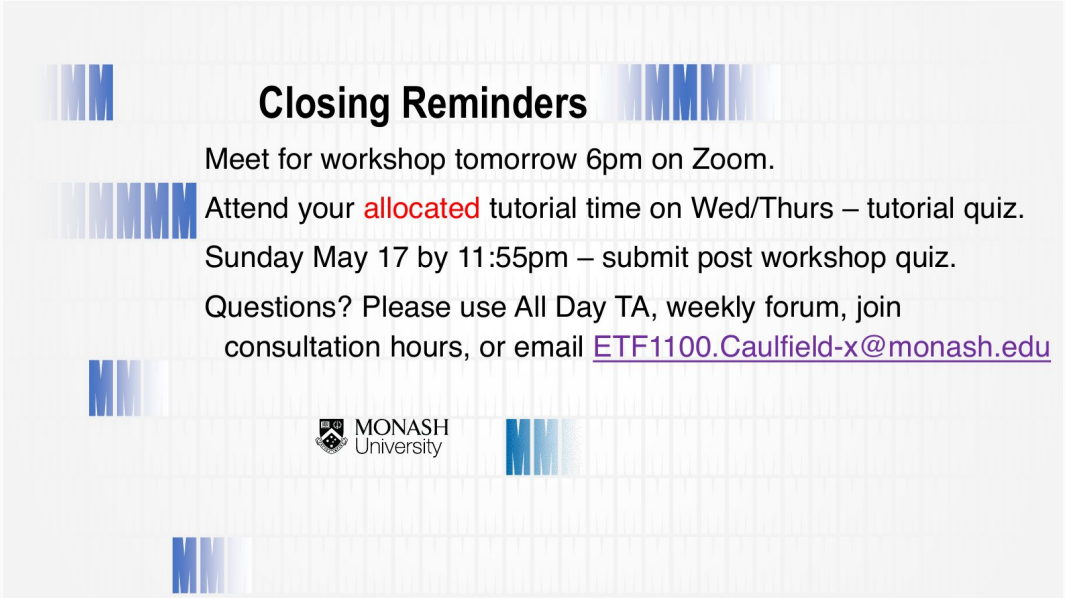


Figure 32: ETF1100 Week 10 Seminar source PDF, page 32